

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools.

Additional Service Items

- Gasket scraper.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

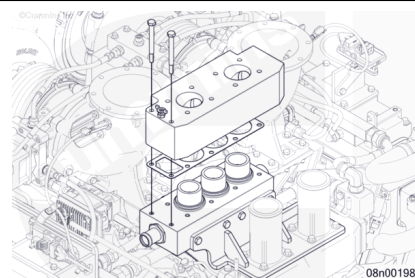
- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-018.html>).
- Disconnect the radiator hoses from the low-temperature aftercooler (LTA) water outlet connection. See equipment manufacturer service information
- Disconnect the aftercooler lines from the LTA thermostat housing. Refer to Procedure 010-006 in Section 10 (</qs3/pubsys2/xml/en/procedures/57/57-010-006-tr.html>).

- Remove the LTA water outlet connection and cover plate. Refer to Procedure 008-110 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-110.html>).

Remove

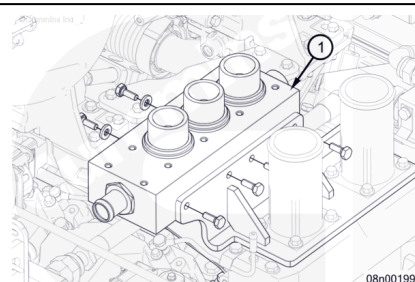
Remove the remaining two capscrews, LTA thermostat housing, and gasket.

Discard the gasket.



Remove the LTA thermostat housing support capscrews from the front and rear brackets.

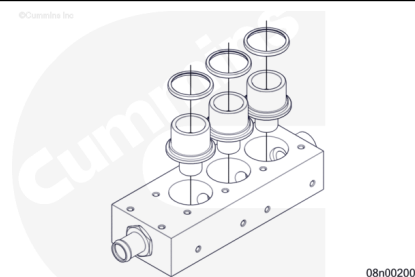
Remove the LTA thermostat housing support (1).



Disassemble

Remove the LTA thermostats and thermostat seals. Refer to Procedure 008-131 in Section 8.

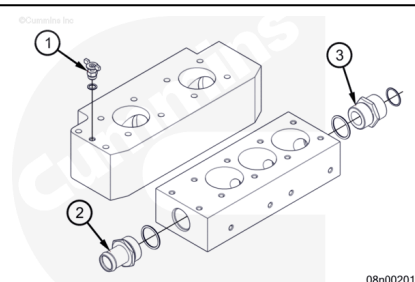
(</qs3/pubsys2/xml/en/procedures/57/57-008-131.html>)



Remove the draincock (1) from the thermostat housing.

Remove the hose coupling (2), male union (3), and o-rings from the LTA thermostat housing support.

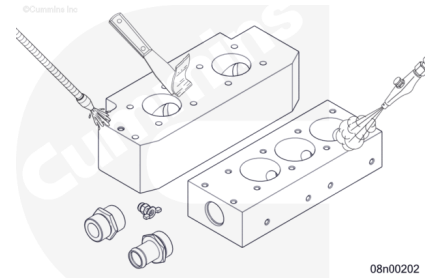
Discard the o-rings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove any gasket material from the LTA thermostat housing and support. Use a gasket scraper.

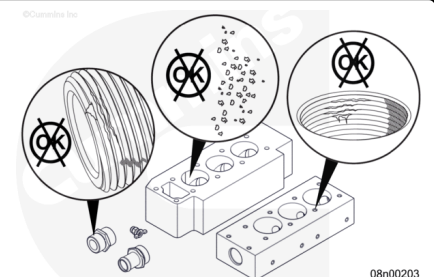
Clean the components with solvent.

Dry with compressed air.

Inspect the thermostat housing for the following:

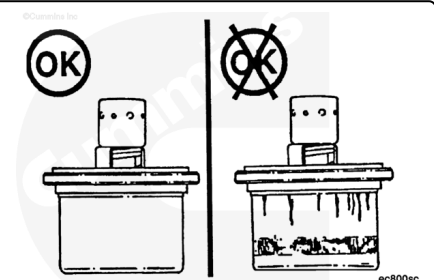
- Cracks
- Thread damage
- Damage to sealing surfaces
- Pitting and corrosion.

Replace any damaged components.



Check the thermostat for wear or damage.

If the barrel of the thermostat is worn or fretted, it **must** be replaced. Refer to Procedure 008-013 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-013-tr.html>).



Assemble

Install the draincock (1) into the thermostat housing.

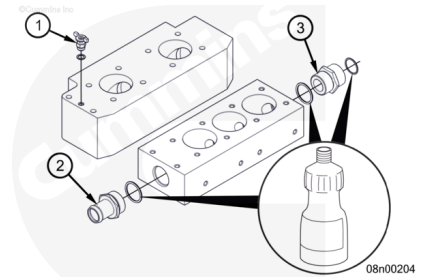
Torque Value: 47 n•m [35 ft-lb]

Install new o-rings on the hose coupling and male union.

Apply vegetable oil to the o-rings.

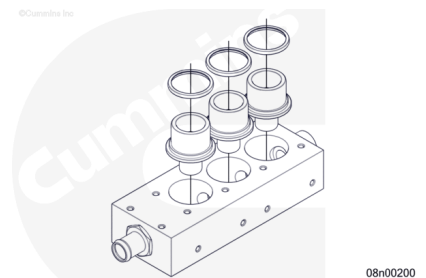
Install the hose coupling (2) and male union (3) into the thermostat housing support.

Torque Value: 68 n•m [50 ft-lb]



Install the LTA thermostat seals. Refer to Procedure 008-165 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-165.html>)

Install the LTA thermostats. Refer to Procedure 008-131 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-131.html>)

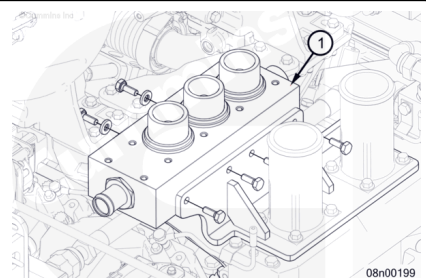


Install

Install the LTA thermostat housing support (1) between the mounting brackets.

Install the capscrews.

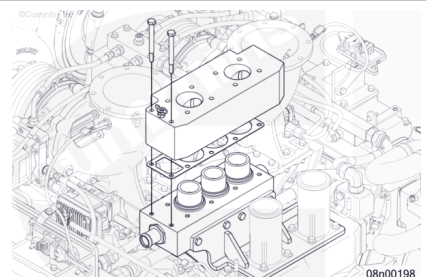
Torque Value: 65 n•m [48 ft-lb]



Install the LTA thermostat housing onto the thermostat housing support. Use a new gasket.

Install the two mounting capscrews on the right bank side of the LTA thermostat housing. Hand tighten **only**.

Note : The LTA thermostat housing capscrews must be tightened with the LTA thermostat housing and LTA water outlet connection capscrews.

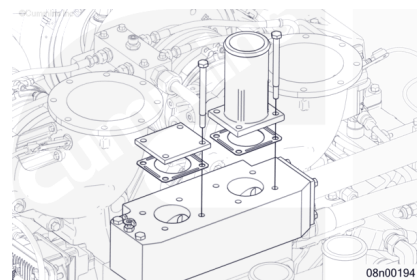


Install the LTA water outlet connection and cover plate. Refer to Procedure 008-110 in Section 8.

(/qs3/pubsys2/xml/en/procedures/57/57-008-110.html)

Tighten the 10 mounting capscrews to secure the LTA thermostat housing.

Torque Value: 65 n•m [48 ft-lb]



Finishing Steps

⚠ WARNING ⚠

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⚠ WARNING ⚠

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- Connect the aftercooler lines to the LTA thermostat housing. Refer to Procedure 010-006 in Section 10 (/qs3/pubsys2/xml/en/procedures/57/57-010-006-tr.html).
- Connect the radiator hose to the LTA water outlet connection. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/57/57-008-018.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 85°C [185°F] coolant temperature. Check for leaks.